Name: Prarabdh Srivastava

Batch:Data Analytics Dec Live Batch

Assignment:Power BI

Question 1 : What is Power BI and why is it used in businesses?

Answer: Power BI is a business intelligence and data analytics tool developed by Microsoft that allows organizations to collect, process, analyze, and visualize data from multiple sources in one place.
It helps convert raw data into meaningful, easy-to-understand visual reports such as charts, graphs, and dashboards.

Power BI is widely used in businesses because it supports data-driven decision-making.
Instead of relying on guesses or manual reports, users can see real-time data insights.
It connects to various data sources like Excel, databases, cloud services, and enterprise systems.

The tool allows employees at different levels to access the same accurate data.
Managers use it to monitor key performance indicators (KPIs).
Executives use dashboards to get a high-level overview of business performance.

Power BI also improves productivity by automating reporting tasks.
Reports can be refreshed automatically when new data is added.
Users do not need advanced technical skills to use Power BI effectively.

Its interactive features allow users to filter and drill down into data.
This helps identify trends, patterns, and potential problems quickly.
Power BI can be accessed on desktops, mobile devices, and web browsers.

Overall, Power BI helps businesses save time, improve accuracy, and make smarter strategic decisions based on reliable data insights.

Question 2 : Name and explain the three main components of Power BI.

Answer: The three main components of Power BI are:

1. Power BI Desktop

Power BI Desktop is a Windows-based application used to create reports.
It allows users to connect to different data sources such as Excel, databases, and online services.
Users can clean, transform, and model data using Power Query.

It is mainly used by data analysts and report creators.
Once reports are created, they can be published to the Power BI Service.

2. Power BI Service

Power BI Service is a cloud-based platform (PowerBI.com).
It is used to publish, share, and manage reports and dashboards online.
Users can collaborate with others and access reports through a web browser.
It enables automatic data refresh and real-time dashboards. Managers and decision-makers mainly use this component.

3. Power BI Mobile

Power BI Mobile is a mobile application available for Android, iOS, and Windows devices.
It allows users to view and interact with reports and dashboards on smartphones and tablets.
Users can monitor business performance anytime and anywhere.
It is useful for executives and field employees who need quick insights on the go.

Together, these three components help businesses create, share, and access data insights efficiently.

Question 3 : Explain the Power BI workflow.

Answer: Power BI Workflow

The Power BI workflow describes the step-by-step process of turning raw data into useful business insights.

1. The workflow begins by connecting to data sources such as Excel files, databases, cloud services, or SQL servers.
2. This data is imported into Power BI Desktop, where most of the work is done.
3. Next, the data is cleaned and transformed using Power Query to remove errors, duplicates, or unwanted data.
4. The cleaned data is then modeled, where relationships between tables are created and calculations are added using DAX.
5. After modeling, users create reports and visualizations like charts, tables, and graphs.
6. These visuals help present data in a clear and meaningful way.
7. Once reports are ready, they are published to the Power BI Service (cloud platform).
8. In the Power BI Service, reports can be shared with colleagues or teams.
9. Dashboards are created to combine key visuals from multiple reports.
10. Data can be set to refresh automatically to keep reports up to date.
11. Users can access reports through web browsers or mobile devices.

12. Finally, decision-makers use these insights to monitor performance and make informed business decisions.

Question 4 : List any four data cleaning tasks that can be performed in Power Query.

Answer: Here are four common data cleaning tasks that can be performed in Power Query:

1. Remove Duplicate Rows
   Eliminates repeated records to ensure data accuracy and avoid double counting.
2. Handle Missing or Null Values
   Replaces, removes, or fills empty/null cells to maintain data consistency.
3. Change Data Types
   Converts columns to the correct data type (e.g., text, number, date) for proper analysis.
4. Remove Unnecessary Columns or Rows
   Deletes irrelevant data to simplify the dataset and improve performance.

These tasks help prepare clean, reliable data for analysis in Power BI.

Question 5 : Write step by step instructions to load the above dataset.

Below are step-by-step instructions to load the given dataset into Power BI Desktop.

Steps to Load the Dataset into Power BI

Step 1: Download the Dataset

1. Open the link: https://drive.google.com/file/d/1U6xdJVJPmDF369v2VomQKiF2U0K2t-vl/view
2. Click on the Download button in Google Drive.
3. Save the file to a known location on your computer (for example: Documents or Desktop).

Step 2: Open Power BI Desktop

1. Launch Power BI Desktop on your computer.
2. When the home screen appears, click on Get Data.

Step 3: Select the Data Source *(The*

option depends on the file type)

- If the file is Excel (.xlsx):

    1. Click Excel → Connect.
- If the file is CSV (.csv):
    1. Click Text/CSV → Connect.

Step 4: Locate and Load the File

1. Browse to the folder where you saved the downloaded dataset.
2. Select the file and click Open.

Step 5: Preview the Data

1. Power BI will display a preview of the dataset.
2. You can:
   - Click Load to directly load the data, or
   - Click Transform Data to clean or modify the data using Power Query.

Step 6: Load Data into Power BI

1. If you selected Load, the data will be imported into Power BI.
2. The dataset will appear in the Fields pane on the right side.

Step 7: Verify the Data

1. Check columns and data values in Data View.
2. Ensure the data has loaded correctly and is ready for analysis.

Question 6 : Define Data View, Report View, and Model View. Explain the purpose of each view.

Answer: Data View, Report View, and Model View in Power BI

Power BI provides three main views, each designed for a specific purpose in the data analysis process.

1. Data View

Data View displays the data in a table format, similar to Excel.

Purpose:

- To view the data after it has been loaded into Power BI
- To check rows, columns, and values for accuracy
- To create calculated columns and measures
- To verify data types and basic data quality

This view helps users understand the structure and content of their data before building reports.

2. Report View

Report View is the main and most commonly used view in Power BI.

-

•

Purpose:

To create interactive visualizations such as charts, tables, maps, and graphs

To design reports using drag-and-drop features

- To apply filters, slicers, and formatting
- To analyze data and present insights in a visual way

This view is used to communicate insights and support decision-making.

3. Model View

Model View shows the relationships between different tables in the dataset.

Purpose:

- To create, edit, and manage relationships between tables
- To define how data from multiple tables connects
- To improve data accuracy and performance
- To build a strong data model for correct analysis

This view helps ensure that reports and visuals work correctly across multiple tables.

Summary

- Data View focuses on data inspection and calculations
- Report View focuses on visual analysis and storytelling
- Model View focuses on relationships and data structure

Together, these views allow users to efficiently build, analyze, and present data in Power BI.

Question 7 : Discuss the different data sources that Power BI supports.

Answer: Different Data Sources Supported by Power BI

Power BI supports a wide range of data sources, allowing businesses to analyze data from multiple systems in one platform. These data sources can be broadly grouped into the following categories:

1. File-Based Data Sources

Power BI can connect to files stored locally or on shared drives.

•

•

Examples:

- Excel (.xlsx)
- CSV / Text files
- XML files JSON files PDF files

Purpose:
Used for importing structured or semi-structured data commonly used in business reporting and analysis.

2. Database Data Sources

Power BI supports connections to various relational and non-relational databases.

Examples:

- SQL Server
- MySQL
- Oracle Database
- PostgreSQL
- IBM DB2
- Microsoft Access

Purpose:
Used to analyze large volumes of enterprise data stored in databases.

3. Cloud-Based Data Sources

Power BI integrates well with cloud platforms and online services.

Examples:

- Azure SQL Database
- Azure Data Warehouse
- Azure Blob Storage
- Google BigQuery
- Amazon Redshift

Purpose:
Allows organizations to analyze scalable cloud data efficiently.

4. Online Services (SaaS Applications)

Power BI can connect directly to popular software-as-a-service platforms.

•

•

Examples:

- SharePoint Online
- Microsoft OneDrive
- Salesforce Dynamics 365

•

- Google Analytics

Purpose:
Used for analyzing business data from online applications without manual exports.

5. Web and API Data Sources

Power BI supports importing data directly from the web.

Examples:

- Web pages (URLs)
- REST APIs
- OData feeds

Purpose:
Useful for accessing publicly available or API-based data.

6. Other Data Sources

Power BI also supports additional specialized connectors.

Examples:

- Hadoop Distributed File System (HDFS)
- Spark
- R and Python scripts

Purpose:
Used for advanced analytics and big data processing.

Question 8 : Split Owner Name to create two new columns as First Name and Last Name

Answer: Step-by-Step Instructions

1. Open Power BI Desktop.
2. Go to the Home tab and click Transform Data to open Power Query Editor.
3. In Power Query, select the column named Owner Name.
4. Go to the Transform tab in the top menu.
5. Click on Split Column.
6. Choose By Delimiter.
7. Select Space as the delimiter (since first and last names are usually separated by a space).

8. Choose Split at Left-most Delimiter (for First Name and Last Name).
9. Click OK.

Result

- The Owner Name column will be split into two new columns.
- Rename them as: ○ First Name ○ Last Name

Final Step

10. Click Close & Apply to load the changes back into Power BI.